Bitcoin Diploma

Content Gap Analysis Report

Full Diploma (v2026.04) vs. Reduced Syllabus

Full Diploma	Reduced Version
10 modules - 124 pages	3 modules - 4 pages
Full instructional content	*Topic headings only*

This report documents the conceptual content removed from the Bitcoin Diploma to produce the reduced syllabus version. The reduced version organises its three modules as: **Module 1 - The Problem**, **Module 2 - The Solution**, and **Module 3 - Using and Securing Bitcoin**. Its topic headings do not map 1-to-1 to the full diploma's module numbers -- for example, the reduced Module 2 - The Solution draws from full diploma Modules 4, 5, and 6. This report therefore compares **by topic**, not by module number, and flags what is present vs. absent for each concept area.

1. Executive Overview

Content Area	Full Diploma	Reduced Version
Module 1-3 topic headings	**Included**	**Included**
All body text & explanations	**Included**	**Removed**
Activities & exercises	**Included**	**Removed**
Examples, analogies & thought experiments	**Included**	**Removed**
Tables & comparison charts	**Included**	**Removed**
Key takeaways & module summaries	**Included**	**Removed**
Discussion questions	**Included**	**Removed**
Types of Money (section 1.5)	**Included**	**Removed**
Psychology of Money (section 1.6)	**Included**	**Removed**
Global Debt & Social Inequality (sec. 4.2)	**Included**	**Removed**
The Lightning Network (module 7)	**Included**	**Removed**
Security & Cryptography (module 8)	**Included**	**Removed**
The UTXO Model (module 8.2)	**Included**	**Removed**
Bitcoin Mining in depth (module 9)	**Included**	**Removed**
The Mempool (module 9.2)	**Included**	**Removed**
Future of Bitcoin (module 10)	**Included**	**Removed**
Real-world case studies (appendix)	**Included**	**Removed**
Glossary - ~80 terms (appendix)	**Included**	**Removed**

2. Module 1 - The Problem

The reduced syllabus lists the correct high-level topics for this module. All of the following topic names appear in both versions -- but the full diploma contains explanatory prose, exercises, and examples that are entirely absent from the reduced outline.

2.1 Topics Present in Both Versions

Topic (Reduced Syllabus)	Full Diploma Section	Status
Discussion questions about money	Module 1.1	**Yes**
Functions of money: MoE, SoV, UoA	Module 1.3	**Yes**
Properties: divisibility, acceptability, fungibility...	Module 1.4	**Yes**
Activity: Are these good forms of money? (scoring)	Module 1.4	**Yes**
Reflection: evolution of money	Modules 2.0-2.1	**Yes**
Activity: Barter	Module 2.1	**Yes**
Reflection: money has no intrinsic value	Module 1.2	**Yes**
Quick history of money (timeline)	Module 2.1	**Yes**
Intro to the fiat system	Modules 3.0-3.2	**Yes**
Fractional reserve banking: explanation + activity	Module 3.2	**Yes**
Inflation / money devaluation	Modules 3.2, 4.1	**Yes**
Who controls fiat: govts, wealthy, banks, central banks	Module 3.2	**Yes**
Future of the fiat system: CBDCs	Module 3.3	**Yes**

2.2 Content in the Full Diploma NOT in the Reduced Version

Definition of Money (Module 1.2)

- Full explanatory prose on money as a social contract; the distinction between money representing value *by which* goods are exchanged vs. the value *for which* they are exchanged
- The chocolate-as-money thought experiment illustrating that any widely accepted good can function as money
- The concept of a transaction as a transfer of value, with examples ranging from buying a sandwich to investing in stocks

Types of Money -- Section 1.5 (Entirely Absent)

- Physical money taxonomy: commodity money (gold, silver, gunpowder), representative money (silver certificates), fiat money (Federal Reserve notes)
- Digital money taxonomy: electronic currencies, payment rails, card networks, digital wallets
- Definitions and distinctions among CBDCs, stablecoins, and cryptocurrencies
- The concept of removing intermediaries to make transactions more efficient and decentralise control

Psychology of Money -- Section 1.6 (Entirely Absent)

- Scarcity applied to all resources: the desert water thought experiment
- Artificial (centralised) vs. natural (decentralised) scarcity -- luxury bags vs. precious metals
- Time preference: high vs. low, illustrated with the $100 now vs. $110 in a year example
- Opportunity cost: definition and application to the desert water scenario
- Activity: Time Preference -- candy choice exercise with class reflection questions

Fiat History Detail (Module 3.1)

- Nixon Shock (1971), Bretton Woods (1944), Executive Order 6102 (1933), creation of the Federal Reserve (1913), and the resulting 96% loss of the dollar's value by 1980 -- present as a detailed narrative and timeline in the full diploma, absent from the reduced version

"Too Big to Fail" (Module 3.2)

- The 2008 financial crisis, Lehman Brothers, AIG, government bailouts, Basel III, and the G-SIBs concept

Monetary and Fiscal Policy Tools (Module 3.2)

- Target rates: unemployment below 6.5%, GDP growth 2-3%, core inflation 2.0-2.5%
- Expansionary vs. contractionary fiscal policy; exchange rate policies and price controls

Global Debt Burden and Social Inequality -- Section 4.2 (Entirely Absent)

This section provides the motivational framing for why Bitcoin was created. It is entirely missing from the reduced version.

- Jaime's story: a college student whose purchasing power erodes year-over-year despite stable income
- Grandfather's price comparisons: $1 buying 10 bags of pretzels in 1913 vs. $9.69 in 2020
- A full monthly ledger example; year-on-year cost table (rent +12.5%, groceries +15%, total +10%)
- The US dollar losing 97% of purchasing power since 1913 (goods costing 30x more)
- Productivity vs. compensation chart 1948-2017: wages stagnating while productivity rises
- Wealth concentration: the rich benefit from inflation via appreciating assets; the poor lose via cash erosion
- The global debt spiral: US federal debt rising from ~$23 trillion (2019) to ~$37 trillion today

3. Module 2 - The Solution

The reduced Module 2 draws content from full diploma Modules 4, 5, and parts of 6. All listed topics are present in the full diploma, but significant depth and several sub-topics are absent from the reduced version.

3.1 Topics Present in Both Versions

Topic (Reduced Syllabus)	Full Diploma Section	Status
Intro to the Cypherpunks	Module 4.3	**Yes**
Centralised vs. decentralised systems	Module 4.3	**Yes**
History of digital currencies: Ecash, Digicash, B-money..	Module 4.3	**Yes**
Common objectives + points of failure (early systems)	Module 4.3	**Yes**
The cryptography mailing list + Bitcoin whitepaper	Module 5.0	**Yes**
Bitcoin history: whitepaper, genesis block, Pizza Day	Module 5.0	**Yes**
Today Bitcoin is: decentralised, digital, P2P, finite	Module 5.2	**Yes**
How Bitcoin works: users, nodes, miners	Module 5.1	**Yes**
Quick explainer of Proof of Work	Module 5.1	**Yes**
21 million limit	Module 5.1	**Yes**
Halving	Modules 5.1, 9.1	**Yes**
Bitcoin divisibility / what are satoshis	Modules 5.2, 7.0	**Yes**
How a transaction works (overview)	Modules 5.1, 6.4	**Yes**
Activity: Let's build a blockchain	Module 5.1	**Yes**
Reflection: does Bitcoin meet money characteristics?	Module 5.2	**Yes**
How accounts work / addresses on the blockchain	Modules 5.2, 6.2	**Yes**
Addresses as lockers, keys analogy	Module 6.2	**Yes**
What are keys / seed phrase	Module 6.2	**Yes**
Tree structure: seed to private to public to address	Module 6.2	**Yes**
Quick intro to wallets (interface, balance, send, receive)	Module 6.2	**Yes**

3.2 Content in the Full Diploma NOT in the Reduced Version

Cypherpunk Detail (Module 4.3)

- Key figures: Eric Hughes, Timothy C. May, John Gilmore; the 1992 Cypherpunk Manifesto
- Phil Zimmermann's PGP (Pretty Good Privacy) -- encrypted email for private communication
- The 'Orwellian future' concept -- why cypherpunks opposed centralised surveillance
- The Tor Network as a working example of a decentralised anonymity system

Digital Currency History -- Depth (Module 4.3)

- The full diploma explains *why* each early system failed and what each contributed. The reduced version lists names only (Ecash, Digicash, etc.) without individual analysis
- David Chaum's specific approach: cryptography for secure, private transactions -- but reliant on a central authority, making it censorable

Nakamoto Consensus Detail (Module 5.1)

- The Monopoly analogy for consensus: all players agree to the same rules or are rejected
- The rule that creating extra bitcoin gets you automatically rejected by every node
- The 4th participant type -- **Developers and Projects** -- absent from the reduced version's '3 core participants'
- The Byzantine Generals Problem and how Bitcoin solves it with Proof of Work
- The 'symphony' of decentralisation: how miners, nodes, users, and developers balance each other

Bitcoin as Sound Money -- Full Analysis (Module 5.2)

- Three-stage evolution of sound money: Store of Value to Medium of Exchange to Unit of Account
- The 'digital gold' framing and Bitcoin's decade-long establishment as a store of value
- Bitcoin vs. Gold vs. US Dollar comparison table across 9 properties (durability, portability, divisibility, fungibility, scarcity, verifiability, established history, censorship resistance, programmability)
- The concept of personal responsibility: no customer service, no recovery -- users must secure their own keys

Acquiring Bitcoin -- Section 6.1 (Entirely Absent)

- Four methods: earning bitcoin, mining, in-person P2P exchange, online P2P exchange
- How escrow works on P2P platforms to protect both parties
- Centralised exchanges: easiest method but requiring KYC identity verification
- Key risk: centralised exchanges can misappropriate funds or sell more bitcoin than they hold -- the fractional reserve parallel

Wallet Types and Selection Criteria (Module 6.2)

- Full comparison table of Online, Mobile, Desktop, and Hardware wallets -- advantages, disadvantages, and example user profile for each
- Open source vs. closed source wallet considerations
- Selection criteria: security, privacy, ease of use, compatibility, fees, reputation, control

"Don't Trust, Verify" -- Section 6.5 (Entirely Absent)

- Core Bitcoin philosophy: always verify independently, never follow claims blindly
- Application to investment schemes, 'next Bitcoin' claims, and promises of quick profits
- Why open-source code matters: community verification over trust in a single leader

The Lightning Network -- Module 7 (Entirely Absent)

- The Lightning Network as a second-layer payment protocol built on top of Bitcoin
- The cafe tab analogy: opening a channel, unlimited off-chain transactions, settling on-chain at the end
- How Lightning routes payments through intermediaries without trusting them
- On-chain vs. Lightning comparison: speed, fees, security, and scalability
- Lightning capacity: millions of transactions/sec vs. Bitcoin's 7/sec vs. Visa's 65,000/sec
- Lightning wallet setup: seed phrase backup, PIN, sending and receiving sats
- Activity: Lightning Relay Race -- students send and receive real sats

Bitcoin in Daily Life -- Module 7.5 (Entirely Absent)

- How credit card payment processing works and its real cost to merchants (typically over 3%)
- BTCPay Server: open-source, free payment processor for online and in-person merchants
- BTCMap.org: open-source map for finding bitcoin-accepting merchants
- Gift cards as transitional tools for stores that don't yet accept bitcoin directly
- Bitcoin Circular Economies: the Arnhem (Netherlands) and Bitcoin Beach (El Zonte, El Salvador) examples

Security through Cryptography -- Module 8.1 (Entirely Absent)

- Public/private key cryptography in depth: how it solves the shared-secret problem
- Digital signatures: proving transaction authorisation using a private key, verifiable by anyone with the public key
- Hash functions: turning any input into a fixed-length unique fingerprint (SHA-256, RIPEMD160)
- The avalanche effect: changing one character in the input produces a completely different hash
- Properties of hash functions: deterministic, pre-image resistant, collision resistant, fast to verify
- Activity: Generate a SHA-256 hash -- hands-on hashing tool

The UTXO Model -- Module 8.2 (Entirely Absent)

- What UTXOs (Unspent Transaction Outputs) are -- the 'bills and coins' of Bitcoin
- The $10-bill analogy: spending a 6 BTC UTXO to send 5 BTC produces 0.99 BTC change (after fees)
- Alice/Bob worked example: step-by-step UTXO creation, spending, and change generation
- Why sharing UTXO information enables transaction tracking

Bitcoin Mining in Depth -- Module 9 (Partially Absent)

The reduced version mentions nodes, miners, and PoW briefly. The full diploma covers all of the following additionally:

- Running a node: installing Bitcoin Core, downloading the blockchain, and how this increases personal sovereignty
- Difficulty adjustment: every 2,016 blocks (~2 weeks) to maintain ~10-minute block times
- Full halving schedule with upcoming halvings: 5th (2028, 1.5625 BTC), 6th (2032, 0.78125 BTC), 7th (2036, 0.390625 BTC)

- The four explicit roles of miners: collect transactions, perform PoW, broadcast block, earn rewards
- The Genesis Block: mined by Satoshi Nakamoto on January 3, 2009, unlocking 50 bitcoin

The Mempool -- Module 9.2 (Entirely Absent)

- The mempool as a 'waiting room' for unconfirmed transactions -- the restaurant queue analogy
- How nodes validate transactions before entering the mempool (preventing double-spending)
- How miners select transactions from the mempool, prioritising by fee level
- Activity: Exploring the Mempool using mempool.space

Full Transaction Lifecycle -- Module 9.3 (Absent)

- 9-step walkthrough (Adam to Gerardo): creation, signing, broadcast, node validation, mempool, mining, block validation, blockchain addition, confirmations
- The complete flow diagram: fee selection, wallet signing, node distribution, miner block template, PoW contest, block validation, distribution

4. Module 3 - Using and Securing Bitcoin

The reduced Module 3 maps primarily to Module 6 of the full diploma.

4.1 Topics Present in Both Versions

Topic (Reduced Syllabus)	Full Diploma Section	Status
Custodial vs. non-custodial wallets (explanation)	Module 6.2	**Yes**
Hot wallets: seed phrase on internet-connected device	Module 6.2	**Yes**
Cold wallets: seed phrase generated and kept offline	Module 6.2	**Yes**
Two routes: hot or cold wallet + testnet practice	Module 6.3	**Yes**

4.2 Content in the Full Diploma NOT in the Reduced Version

"Not Your Keys, Not Your Coins" (Module 6.2)

- The full diploma makes this a named, explicit principle with full explanation. The reduced version implies it but never states it

Wallet Setup Detail -- Section 6.3

- Step-by-step instructions: downloading a wallet, generating a seed phrase, confirming the phrase, receiving a first address
- Option 2: restoring an existing wallet from a seed phrase -- a full alternative activity

Receiving and Sending Transactions -- Section 6.4 (Substantially Absent)

- How to share a public address: copy-paste, shareable link, and QR code
- How to send: step-by-step with explicit warnings about irreversibility and double-checking addresses
- The Jim-and-Eliana transaction walkthrough: creating, signing, broadcasting, verifying, and confirming
- Confirmation times: ~10 minutes per confirmation, 2 confirmations recommended for security
- Activity: Transactions in Action -- a four-role exercise (sender, recipient, miner, node operator)

Future of Bitcoin -- Module 10 (Entirely Absent)

- CBDC vs. Bitcoin comparison table: supply, centralisation, monetary policy transparency, permissioning, seizure risk, privacy, and censorship resistance
- 134 countries (98% of global GDP) exploring CBDCs; 66 in advanced phases
- The philosophy of Bitcoin: power distributed equally regardless of wealth, inverting the fiat power dynamic
- Hyperbitcoinization concept
- Benefits of adoption: self-sovereignty, reliable store of value, changes to monetary policy, enhanced transparency, remittance revolution, abundant energy from surplus renewables

- Final reflection activity: revisiting Module 1 questions to measure perspective change

5. Removed Appendices

The following appendices appear in the full diploma and have no equivalent in the reduced version.

5.1 Real-World Case Studies

- **Bitcoin Indonesia:** Diana's story of founding the movement, the Bitcoin House Bali, graduating 200+ students across Bali, Bandung, and Surabaya, and creating a local circular economy where merchants -- including a scooter mechanic and a Jamu drink seller -- accept bitcoin
- **The Core, Kenya:** Felix Mukungu's journey from a crypto scam to building a WhatsApp-based curriculum that grew into a global online platform with 500+ students across Africa, spawning 10+ new projects including Bitcoin Haiti and Kabul Bitcoin

5.2 Additional Resources

- Why Use Bitcoin: 3 curated articles and videos (Boyapati, Svetski, Wiz)
- What Is Bitcoin: 3 curated videos and articles (CuriousInventor, Greg Walker, RT)
- Further Learning: audiobooks, lectures, and the BTC Sessions YouTube channel
- Courses: Summer of Bitcoin, Chaincode Labs, Saylor Academy
- 10 recommended authors: Gladstein, Posch, Booth, Song, Alden, Bhatia, and others
- 15+ curated links: Bitcoin.org, Mempool.space, BTCMap.org, NOSTR, SimpleX, and more

5.3 Key Concepts Reference

- A module-by-module summary of all 10 modules' key learning objectives -- a structured review guide for educators and students

5.4 Glossary

- ~80 defined terms, from '51% Attack' and 'Atomic Swap' through 'UTXO Set', 'Whitepaper', and 'XBT/BTC'
- Each entry includes a precise definition written for beginner-to-intermediate audiences